

Single slit diffraction pattern

1 Aim

- To study the diffraction pattern of a slit using a laser source and a photo-diode and to measure the intensity distribution due to the single slit and measure the slit width.
- To observe the diffraction pattern from an obstacle(thin wire) and measure the thickness of the obstructing wire.

2 Introduction

This experiment is designed to directly view bending of light around an obstacle. The intensity of the diffracted wave is given by

$$I(\theta) = I_0 \left(\frac{\sin(\beta/2)}{\beta/2} \right)^2,$$

where I_0 is the intensity at $\theta = 0$ and $\beta = 2\pi a \sin \theta / \lambda$. Here, a is the width of the slit and θ is the angle the fringe position makes with the central axis.

3 Apparatus in the lab and procedure

1. Switch on the laser source and leave on for around 15 minutes so that the light intensity from the laser source is constant and does not flicker.
2. Mount the slit and adjust its width so that you observe a diffraction pattern on the white screen.
3. Mount the photocell on the optical bench, as far away from the slit as possible. Make suitable adjustments so that the light falls on the small aperture in the photocell.

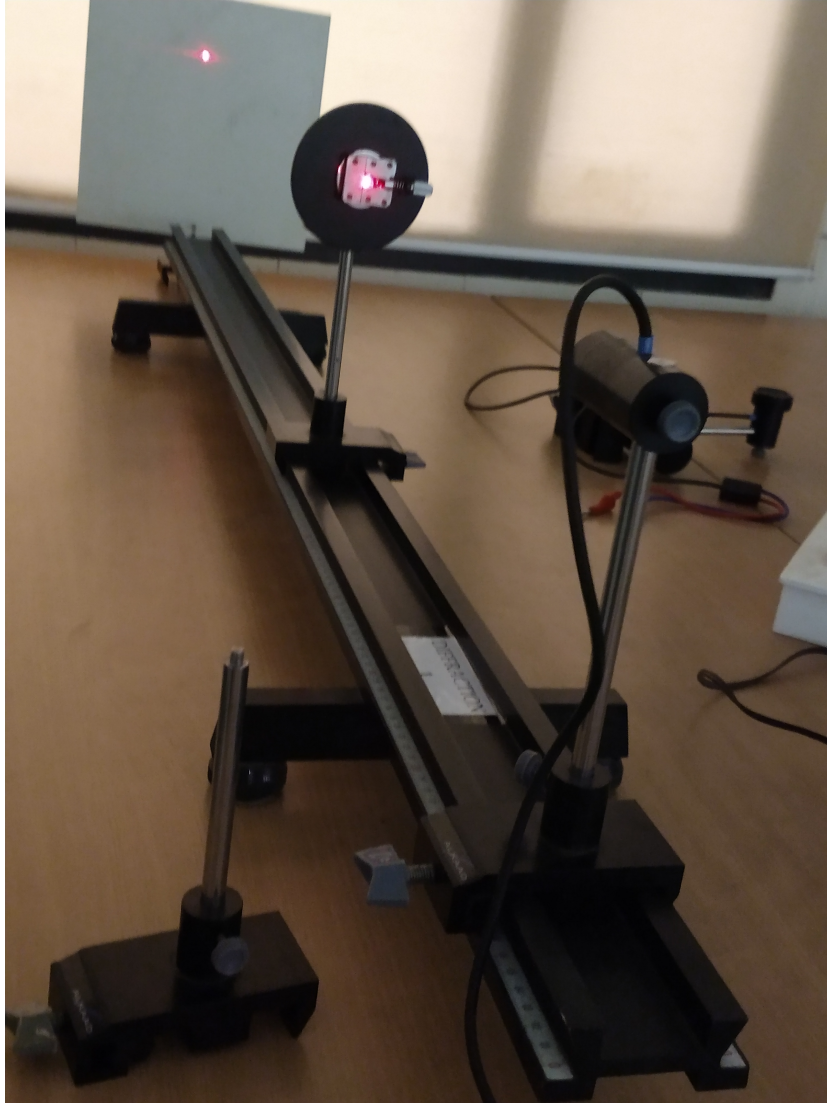


Figure 1: Apparatus in the lab.

4. The photocurrent is proportional to the intensity of the incident light. Measure the photocurrent using the multimeter. Move the photocell using the screws provided, till the entire diffraction pattern is scanned.
5. Plot current as a function of position of the photocell. Measure the distance D between the slit and the photocell and distance x between the centre of the diffraction pattern to the first minimum and compute the fringe width.
6. Adjust the distances between the laser, slit and screen to different lengths and again record the diffraction patterns in each case, and repeat the experiment. Also change the width of the slit and record your observations.
7. In the second part of the experiment, mount a thin wire onto the optical bench. Move the wire and laser source, till you observe the diffraction pattern from the wire. Record the pattern (and explain what you see) in your lab notebook. Measure the obstacle (wire) dimensions directly with a traveling microscope. Calculate the object dimensions from the information in the diffraction pattern. Compare your calculation with the actual dimensions measured directly.

4 Precautions

1. **DO NOT** look at the laser directly.
2. Move the telescope scale only in one direction to avoid back lash error.
3. What if white light is used for this experiment?

4.1 Points to Ponder

1. What kind of pattern would you see if the slit was circular?
2. Find the width of human hair using the above method.